

Overall Methodology Architecture

STAGE 1: DATA PREPARATION

Mahānāma Gold NER (66,960 sentences) + Oversampling (LOC×10, MISC×10)

DCS Sembank Silver NER (12,955 examples from 120 texts) — Novel Extraction Method

Linguistic Regularisation Data (25,627 examples, 15% of total)

Total Training Data: 170,817 examples (74% Gold NER + 11% Silver NER + 15% Linguistic)



STAGE 2: TRAINING (QLoRA + Multi-task)

Base: ByT5-Sanskrit (594M) | QLoRA: Rank=16, Alpha=32, 4-bit NF4

Strategy 1: Full DCS Sembank Silver | Strategy 2: 15% Linguistic (S+L) | Strategy 4: LR=1e-4, 10 epochs

Trainable: 10,764,288 params (1.817%) | Hardware: RTX PRO 4500 Blackwell (34.2 GB)

Training Time: ~17.2 hours | Seed = 42



STAGE 3: EVALUATION (D1/D2/D3)

D1 (In-Domain): 6,672 sentences, Gold NER, Bootstrap 95% CI + McNemar

D2 (Linguistic): 200 sentences, S/L/LM/SLM tasks (Final NER: S=85%, L=93%)

D3 (Cross-Domain): 230 sentences (Pañcatantra), Binary PROPEN (Final NER: 73.4% recall)

Key Trade-off: -0.8 F1 for +16 pp cross-domain + preserved D2 capability



FINAL NER MODEL (F1 = 0.852)